Chapter 3.



SCHOOL OF COMPUTING & DATA SCIENCE
The University of Hong Kong

IDEs and Tools for Mobile App Development

2026-2027
COMP3330 Interactive Mobile Application Design and Programming
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Why IDEs?

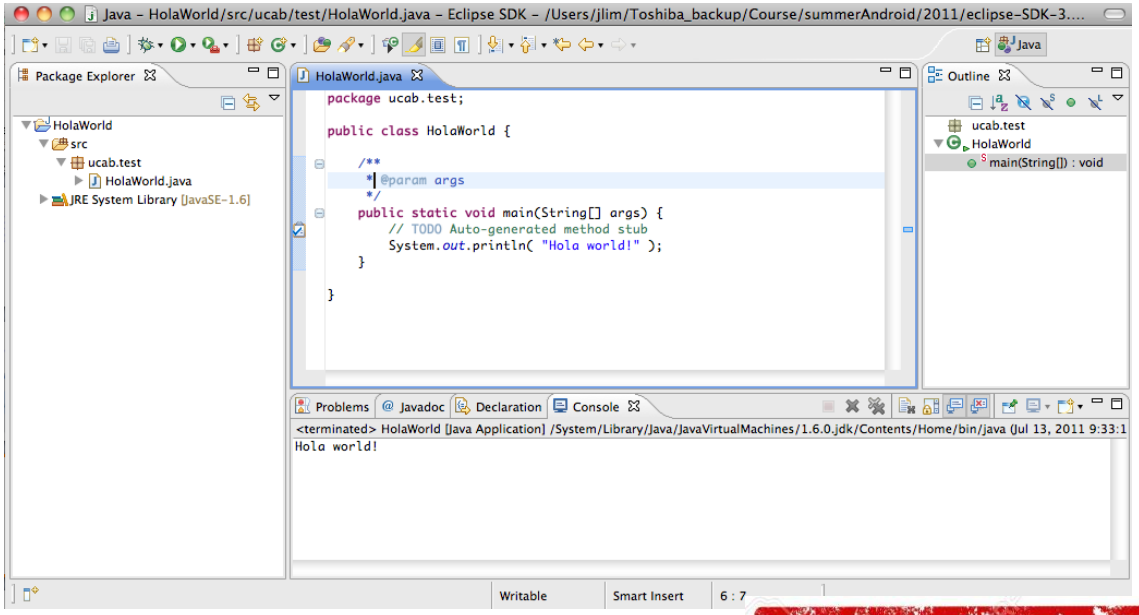
- IDE = Integrated Development Environment
- You cannot write programs on a phone.
  - But you may do so on a tablet
- You need an IDE (to be run on a computer).
  - Software application that allows the computer programmer to develop software for a certain platform on a PC.
  - Normally has: source editor, compiler and/or interpreter and debugger.
  - Upon everything is fine, you can upload / transfer your apps to the smart phone.
  - SDK for a certain model is sometimes required (e.g., Nokia S60)

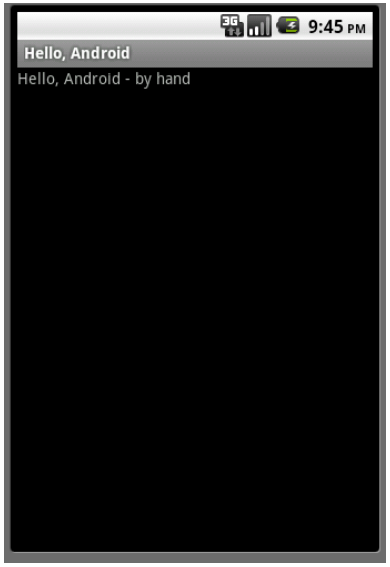
Some tools for mobile apps development

- Android
  - AIDE
  - App Inventor
  - DroidScript
  - CppDroid
  - Android Web Developer (AWD)
  - Python Suite
  - Java Suite
  - Eclipse
  - Arduino
  - Visual Studio
  - Unreal Engine
  - Android Studio
  - Corona
  - PhoneGap
  - …
- iOS
  - Xcode
  - Appcode
  - Atom
  - SublimeText 3
  - CodeRunner 2
  - …

IDE for Android apps - Eclipse

- IDE: Eclipse
- Programming language: Java

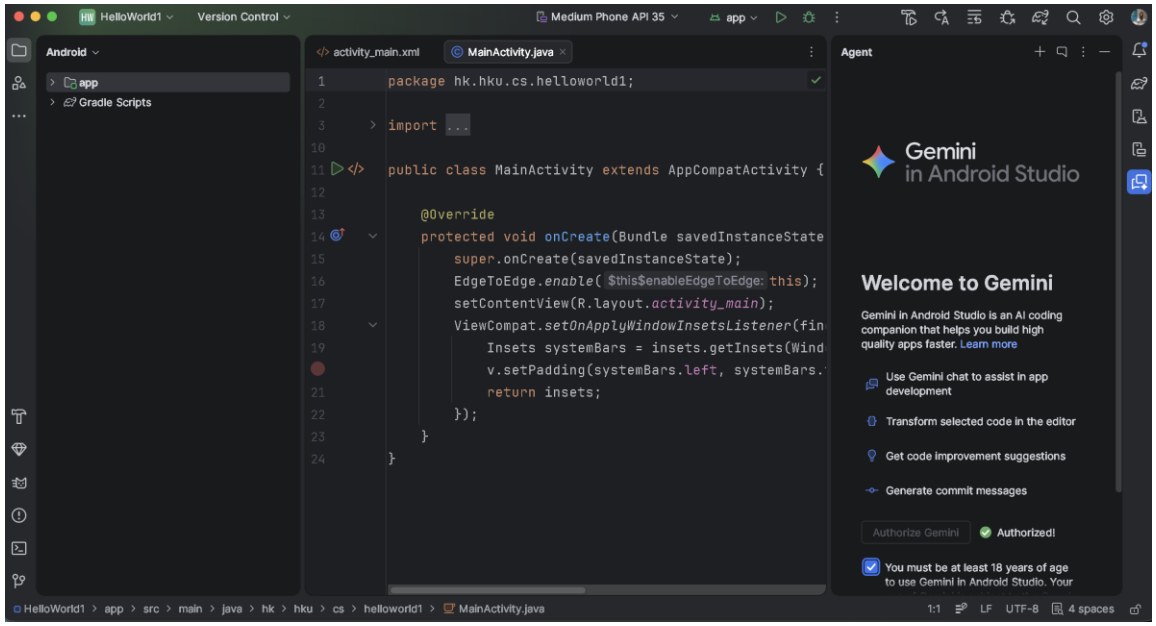


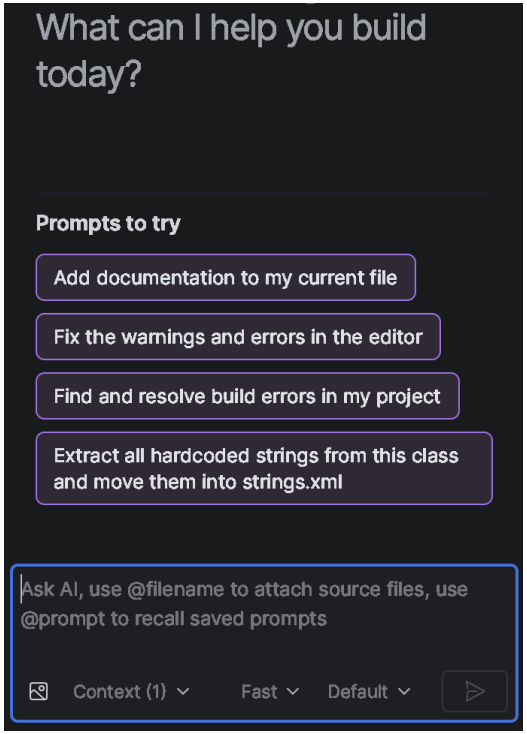




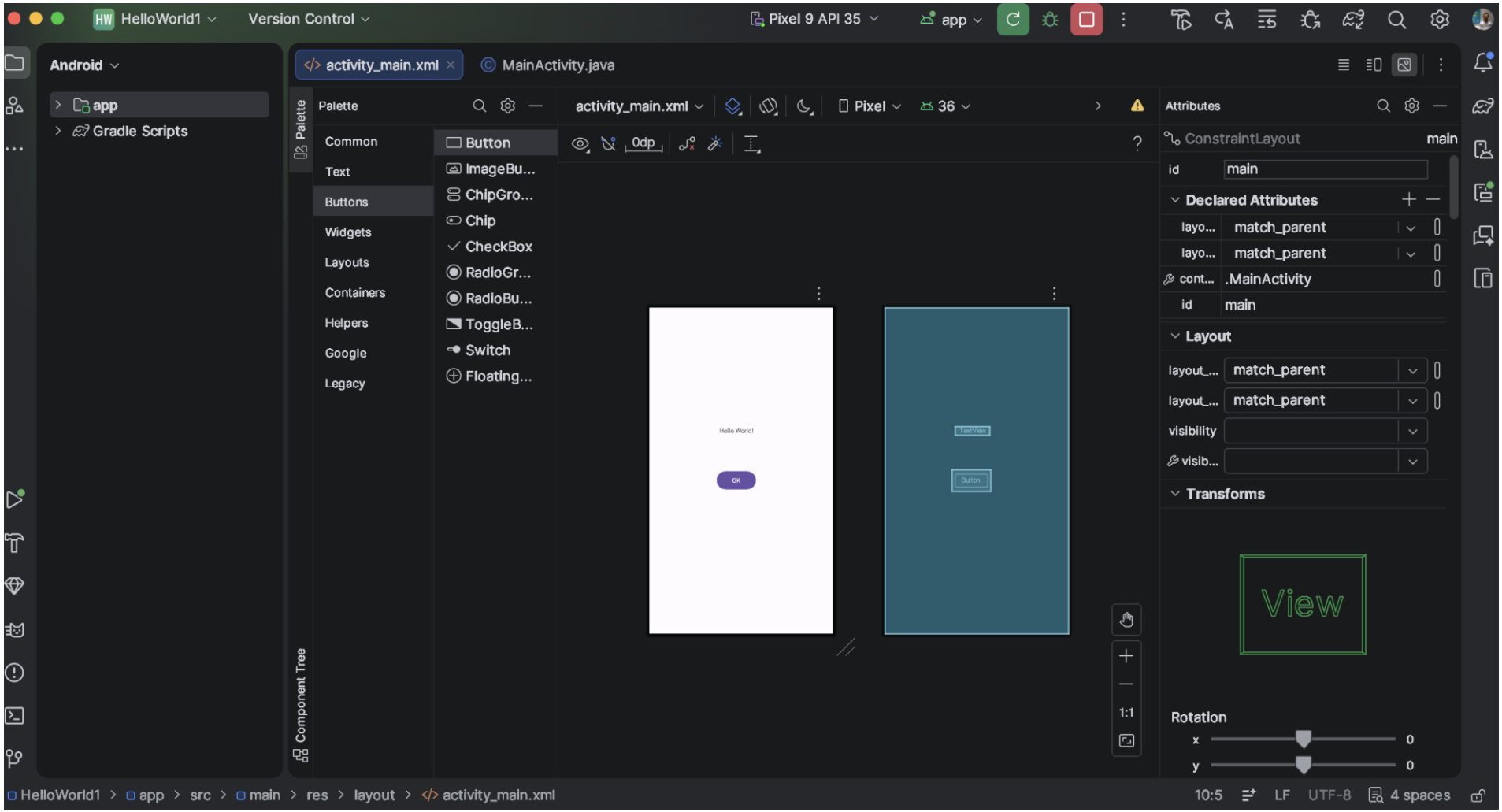
IDE for Android apps – Android Studio

- IDE: Android Studio
- Programming language: Java / Kotlin (aim at simplifying Java syntax)

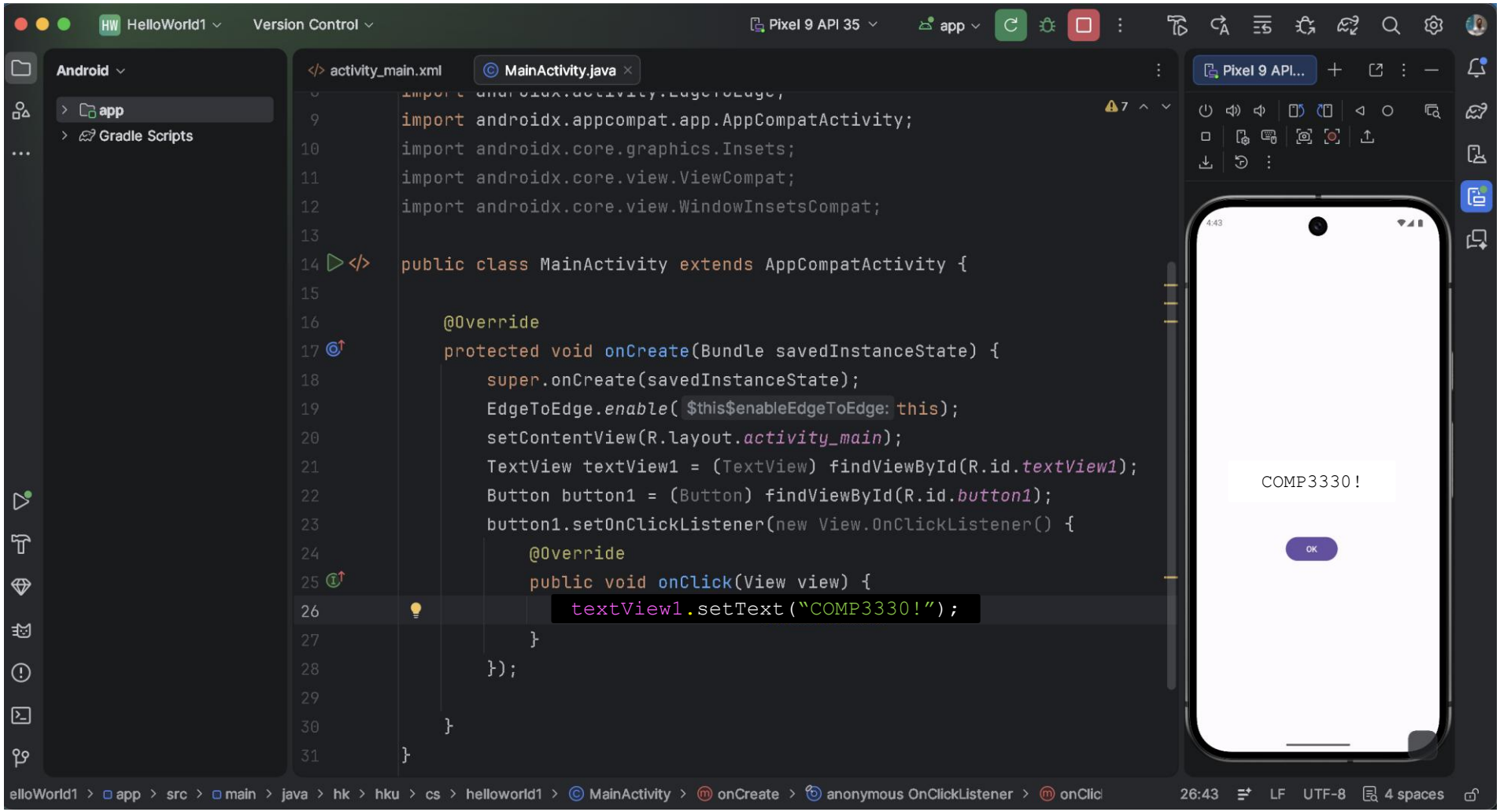




Android Studio

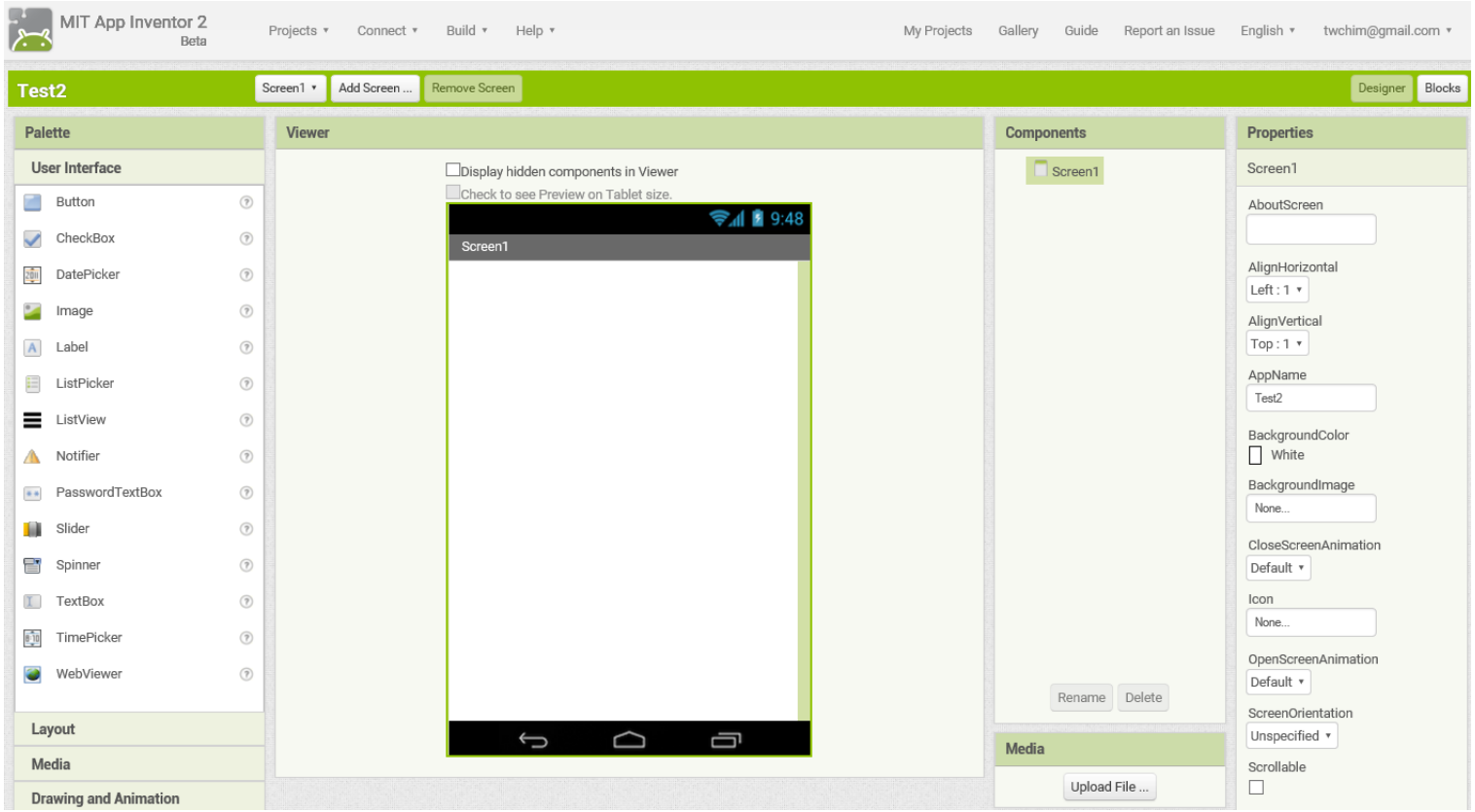


Android Studio



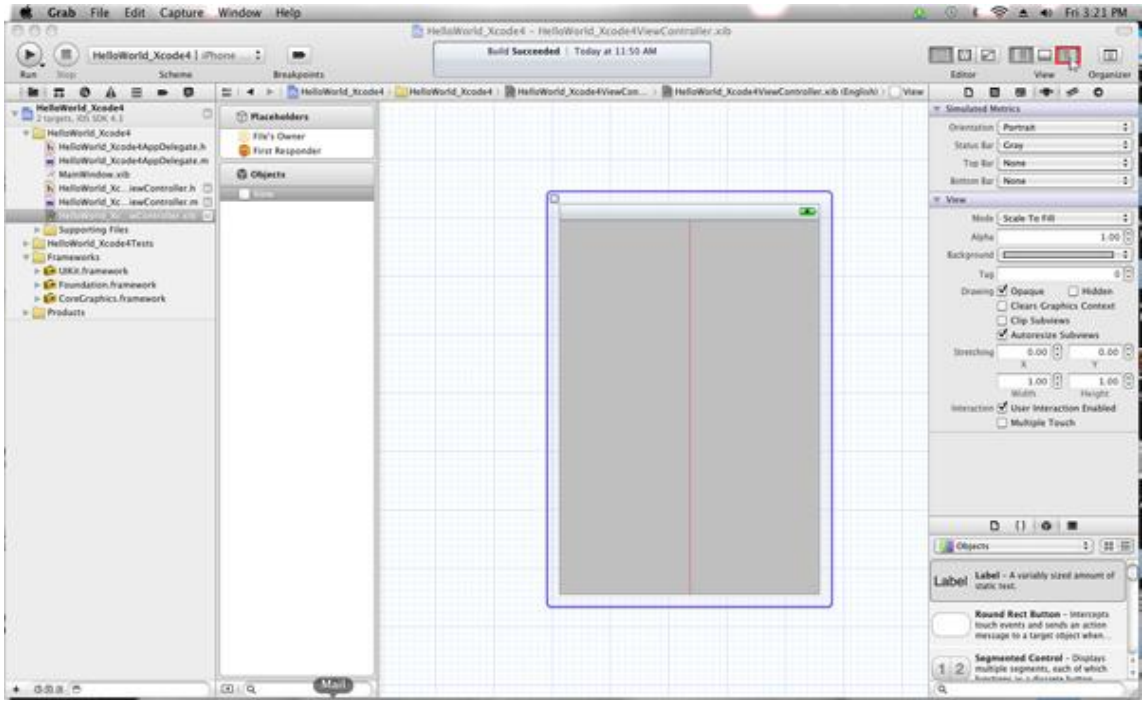
IDE for Android apps – MIT App Inventor

- IDE: MIT App Inventor
- A graphical web-based app development tool for Android



IDE for iOS apps – Xcode

- IDE: Xcode
- Programming language: Objective C / Swift



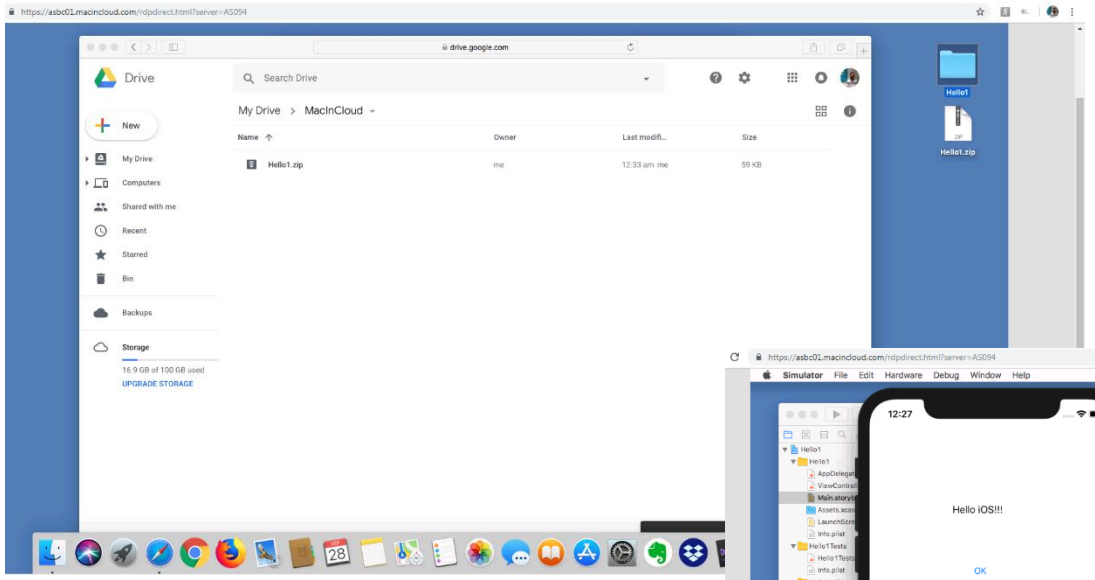


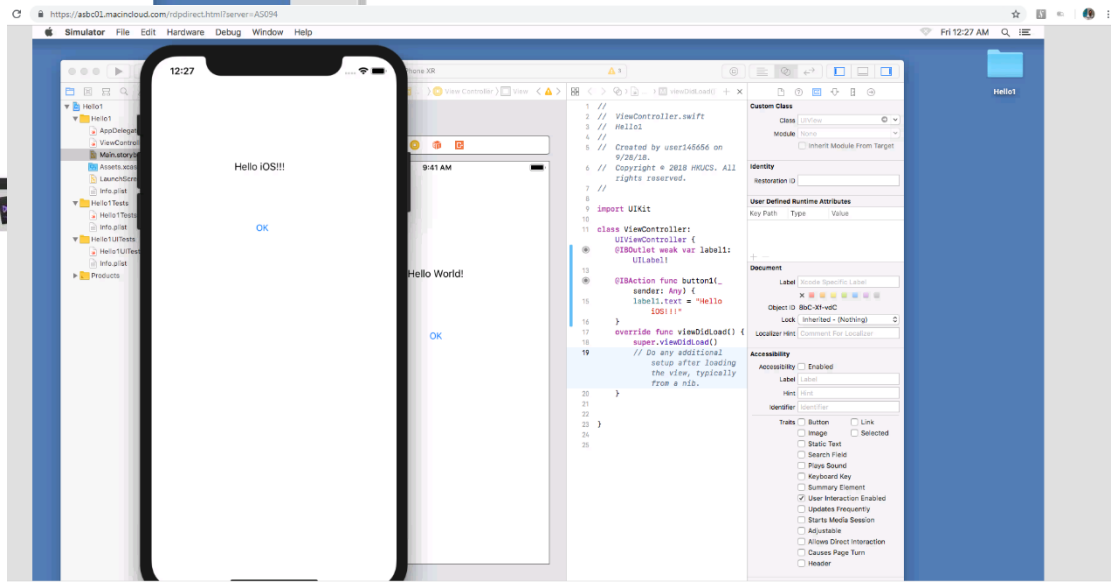
Mac and Xcode

- Xcode can only run on Mac OS.
- If you want to use Mac but do not want to buy a MacBook, you can consider using cloud-based Mac service.
  - MacInCloud (https://checkout.macincloud.com/select)
  - XCodeClub (http://www.xcodeclub.com/)
  - VirtualMacOSX (http://virtualmacosx.com/index.php/shared-plans)
- You can use an ordinary browser to access their site and the virtual Mac environment will appear in form of remote desktop.
- You need to pay for the service. Let's use MacInCloud as an example, you need to pay $1 USD hourly / $4 USD daily / $25 USD monthly plus $0.99 USD initial setup fee.
- All the virtual Mac environment has common software like Xcode installed.

Mac and Xcode

- MacInCloud:





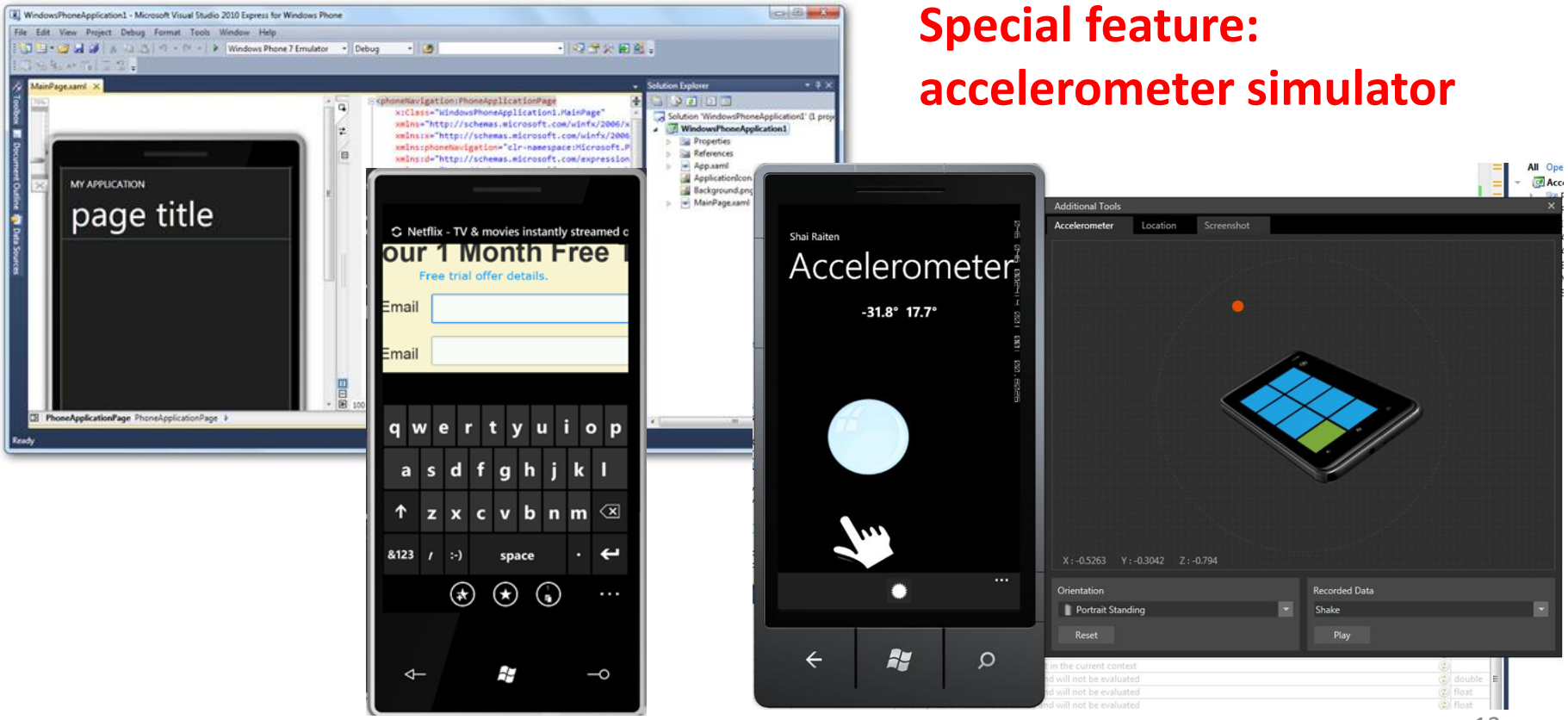
Academic Plan for MacInCloud

- Students can sign up through the academic plan link to enjoy the academic pricing: https://checkout.macincloud.com/select/managed?academic=1
- You are required to provide a picture proof of your active academic status (such as a student card).
- Please refer to the page below for more information regarding the academic plan: https://www.macincloud.com/pages/academic.html

IDE for Windows Phone apps

OUTDATED

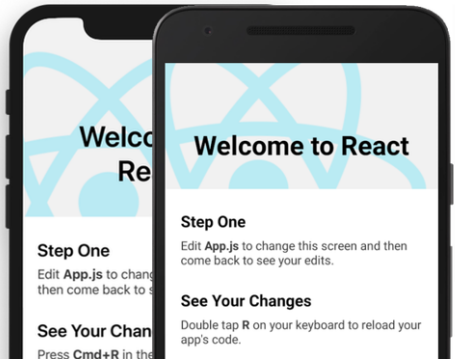
- IDE: Microsoft XNA (an example)
- Programming language: C#



Special feature: accelerometer simulator

Cross-platform Development Tool: React Native





Create native apps for Android and iOS using React

React Native combines the best parts of native development with React, a best-in-class JavaScript library for building user interfaces.

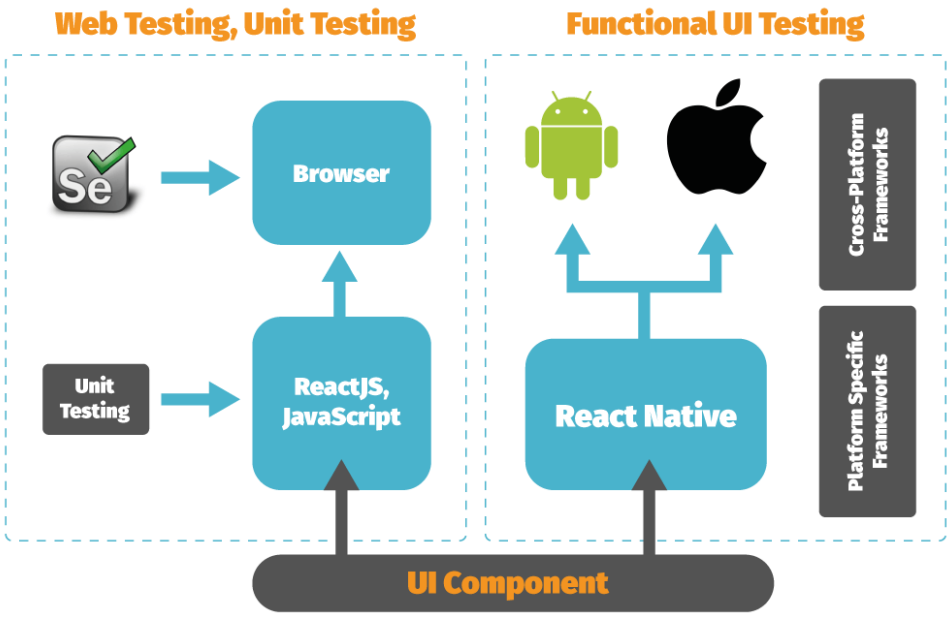
Use a little—or a lot. You can use React Native today in your existing Android and iOS projects or you can create a whole new app from scratch.

```
import React from 'react';
import {Text, View} from 'react-native';
import {Header} from './Header';

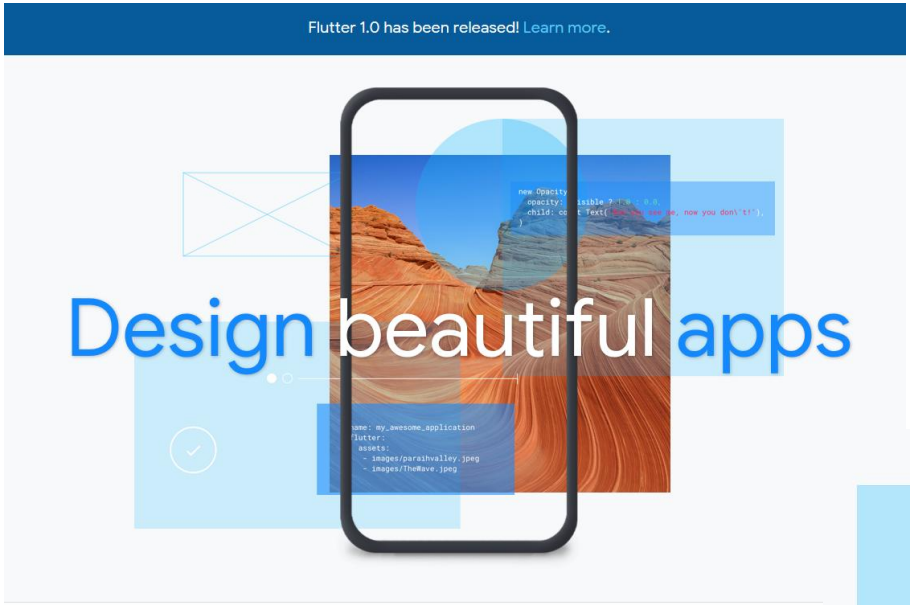
const WelcomeScreen = () =>
  <View>
    <Header title="Welcome to React Native"/>
    <Text style={header}>Step One</Text>
    <Text>
      Edit App.js to change this screen and turn it
      into your app.
    </Text>
    <Text style={header}>See Your Changes</Text>
    <Text>
      Press Cmd + R inside the simulator to reload
      your app's code.
    </Text>
    <Text style={header}>Debug</Text>
    <Text>
      Press Cmd + M or Shake your device to open the
      React Native Debug Menu.
```

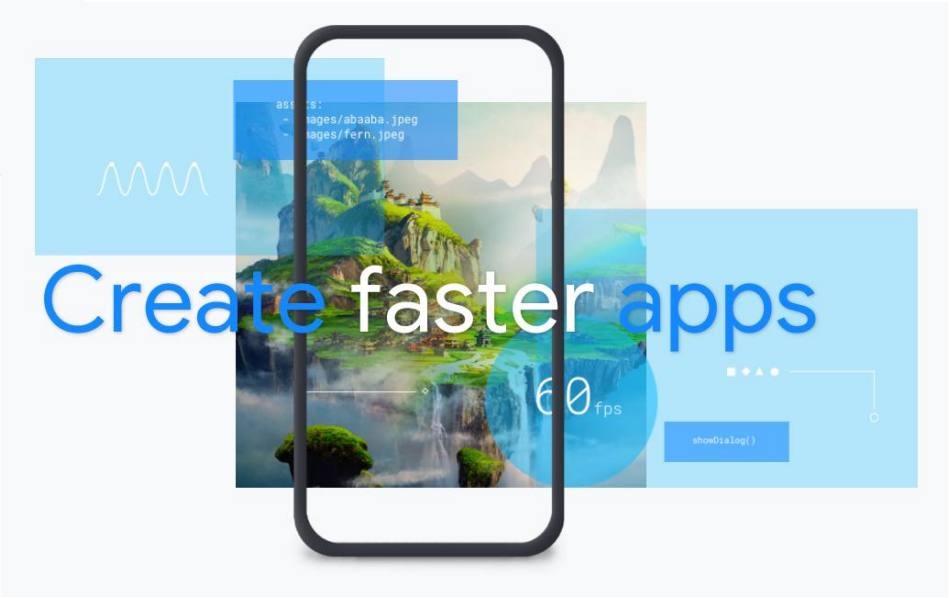
Cross-platform Development Tool: React Native

- Link: https://facebook.github.io/react-native/
- React Native lets you build cross-platform mobile apps using only JavaScript.
- It uses the same design as React (for web applications), letting you compose a rich mobile UI from declarative components.
- You build a real mobile app that's indistinguishable from an app built using Objective-C / Swift / Java / Kotlin.
- React Native uses the same fundamental UI building blocks as regular iOS and Android apps. You just put those building blocks together using JavaScript and React.
- Thousands of apps are using React Native, from established Fortune 500 companies to hot new startups.



Cross-platform Development Tool: Flutter



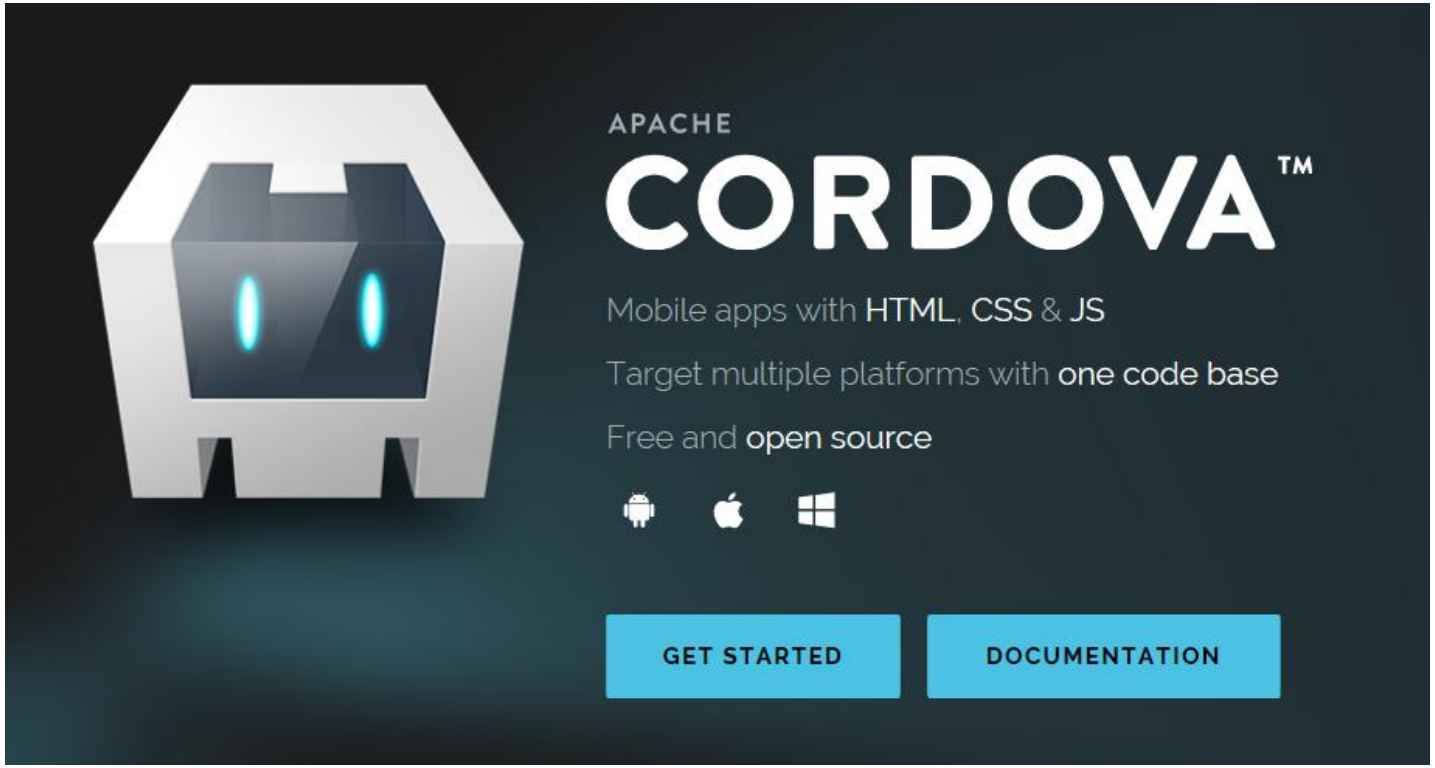


Cross-platform Development Tool: Flutter

- Link: https://flutter.dev/
- Flutter was introduced in May 2017 by Google.
- Flutter is an open-source mobile application development SDK created by Google. It is used to develop applications for Android and iOS.
- Flutter's engine, written primarily in C++, provides low-level rendering support using Google's Skia graphics library. Additionally, it interfaces with platform-specific SDKs such as those provided by Android and iOS.
- Flutter apps are written in the Dart language and make use of many of the language's more advanced features.

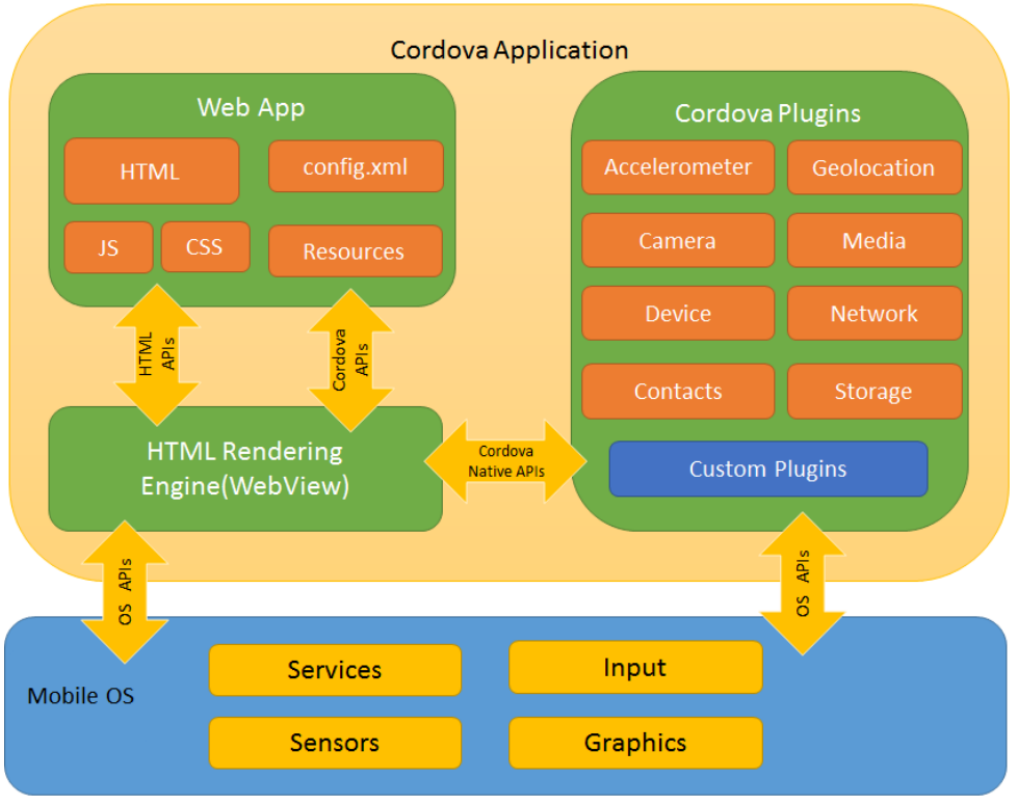


Cross-platform Development Tool: Apache Cordova



Cross-platform Development Tool: Apache Cordova

- Link: https://cordova.apache.org/
- Apache Cordova is an open-source mobile development framework.
- It allows you to use standard web technologies - HTML5, CSS3, and JavaScript for cross-platform development.
- Applications execute within wrappers targeted to each platform, and rely on standards-compliant API bindings to access each device's capabilities such as sensors, data, network status, etc.



Game Engine

- A software framework designed for the creation and development of video games.
- Key components:
  - **Main game program**
    - The actual game logic being implemented using algorithms.
  - **Rendering engine**
    - How to display or project 2D or 3D graphics onto the screen?
  - **Audio engine**
    - Componentry consisting of any algorithm related to sound
  - **Physics engine**
    - Responsible for giving the application a realistic sense of the laws of physics in the application
  - **Artificial intelligence**
    - Usually outsourced from the main game program into a special module to be designed and written by software engineers with specialist knowledge.

Concept of Frame in Game Engine



- User interface is updated frame by frame (whole screen).
- All updates (e.g., motion of characters) are made in a buffer first and then contents of the buffer are copied to the screen (a complete frame).
- As a game usually involves the update of several components, frame concept can ensure smooth animation.
- More details:
  - The time axis is divided into frames and the frames will be updated at pre-defined intervals.
  - Like a cinema movie.
  - If you want some updates, include them into the next frame.

Cross-platform Development Tool: Cocos



Cross-platform Development Tool: Cocos

- Link: https://www.cocos.com/en
- Cocos Creator is both an open source cross-platform 2D & 3D graphics engine and a real-time interactive 2D & 3D digital content creation platform.
- Supports two script languages – JavaScript and TypeScript.
- Related frameworks:
  - Cocos2d-x (2010 – 2019)
  - Cocos Creator 3D (released in 2017)
  - Cocos Creator 2.x (released in 2018)
  - Cocos Creator 3.x (released in early 2021)

Cross-platform Development IDE: Unity (Global) / Tuanjie (China & HKSAR)

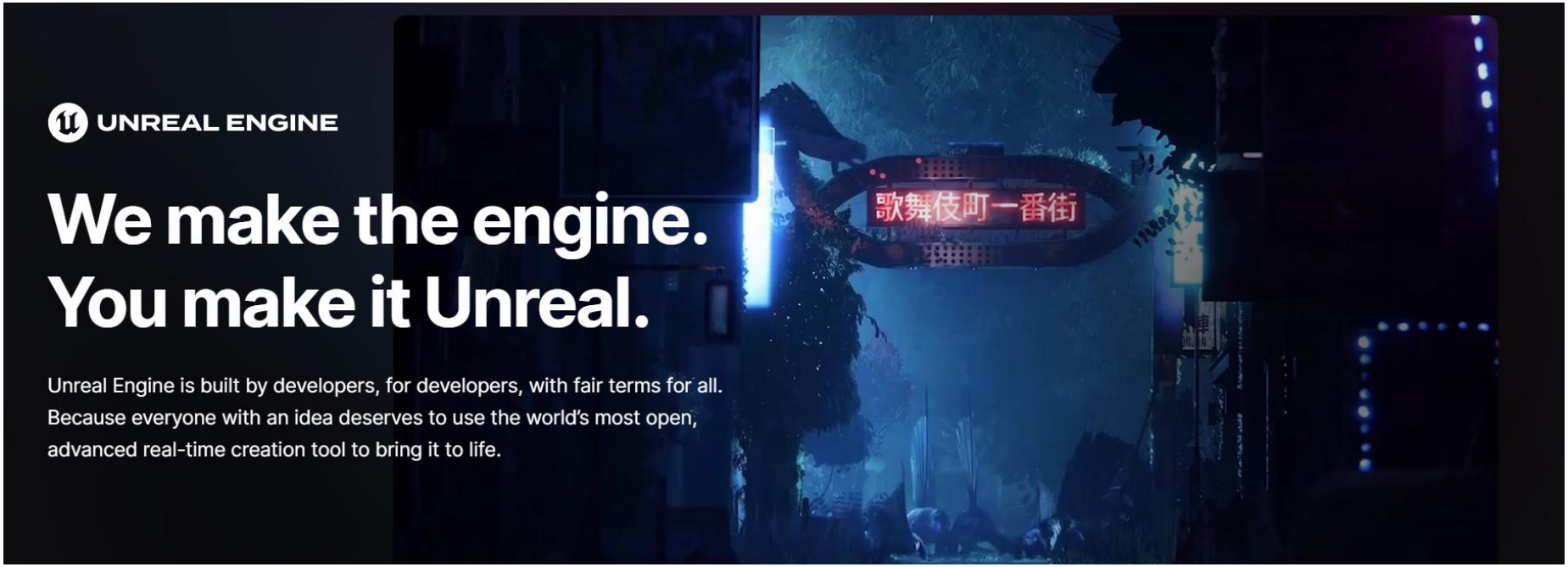


Cross-platform Development IDE: Unity

- Link: http://unity3d.com/ (global) / https://unity.cn/tuanjie/tuanjieying (China & HKSAR)
- Unity / Tuanjie is a popular cross-platform game engine and IDE developed by Unity Technologies, targeting web plugins, desktop platforms and mobile devices.
- Programming language: C#



Cross-platform Development IDE: Unreal



- Link: https://www.unrealengine.com/
- Another popular cross-platform game engine
- Programming language: C++

Mini Program

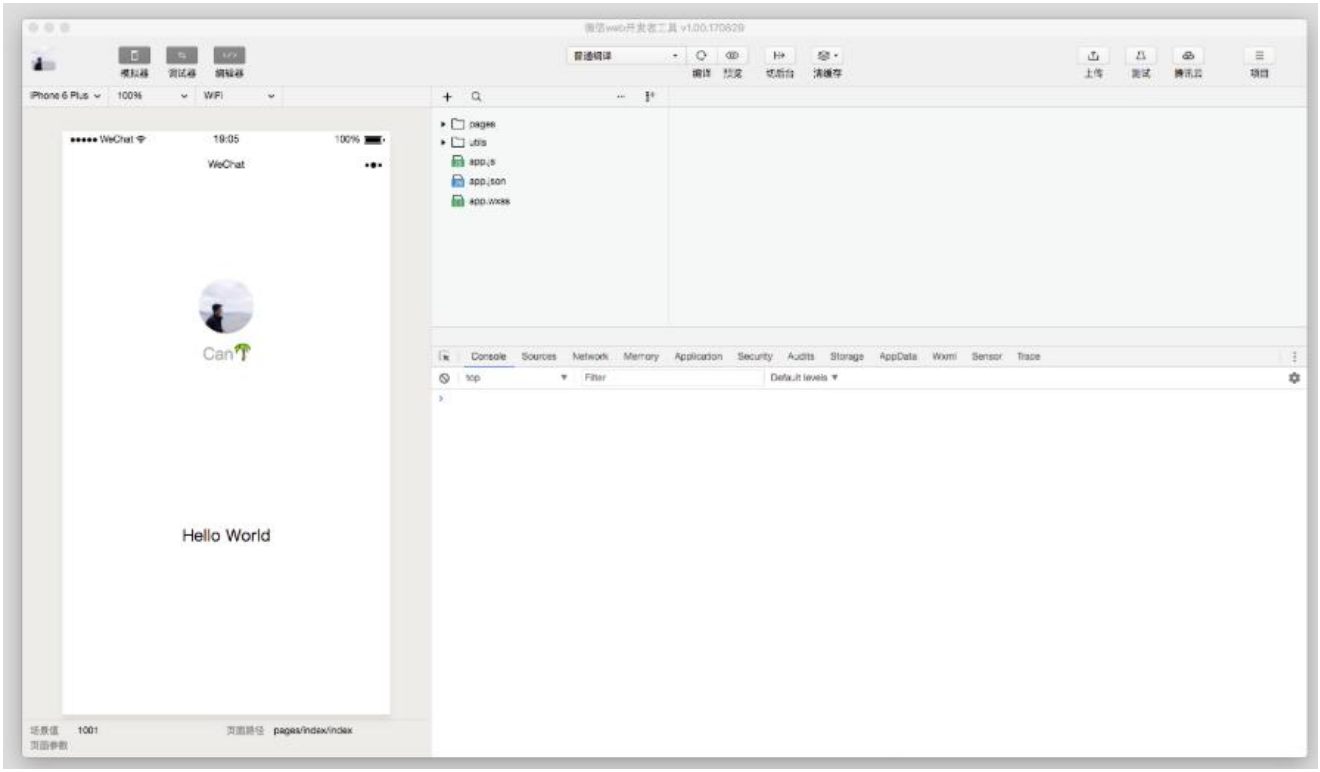
- After WeChat Mini Program was launched at the end of 2016, many similar lightweight programs become popular in China (e.g., Alipay Mini Program, Baidu Mini Program).
- Mini Program is different from traditional mobile applications.
  - It can be obtained directly through a search in another application (e.g., WeChat).
  - It can be obtained by scanning a QR code or via a link shared by another user
  - It occupies very little memory and storage space. No installation is required.
  - It can be opened quickly.
  - It can achieve a user experience closer to the native APP. The set of core functions are nearly the same.

Mini Program

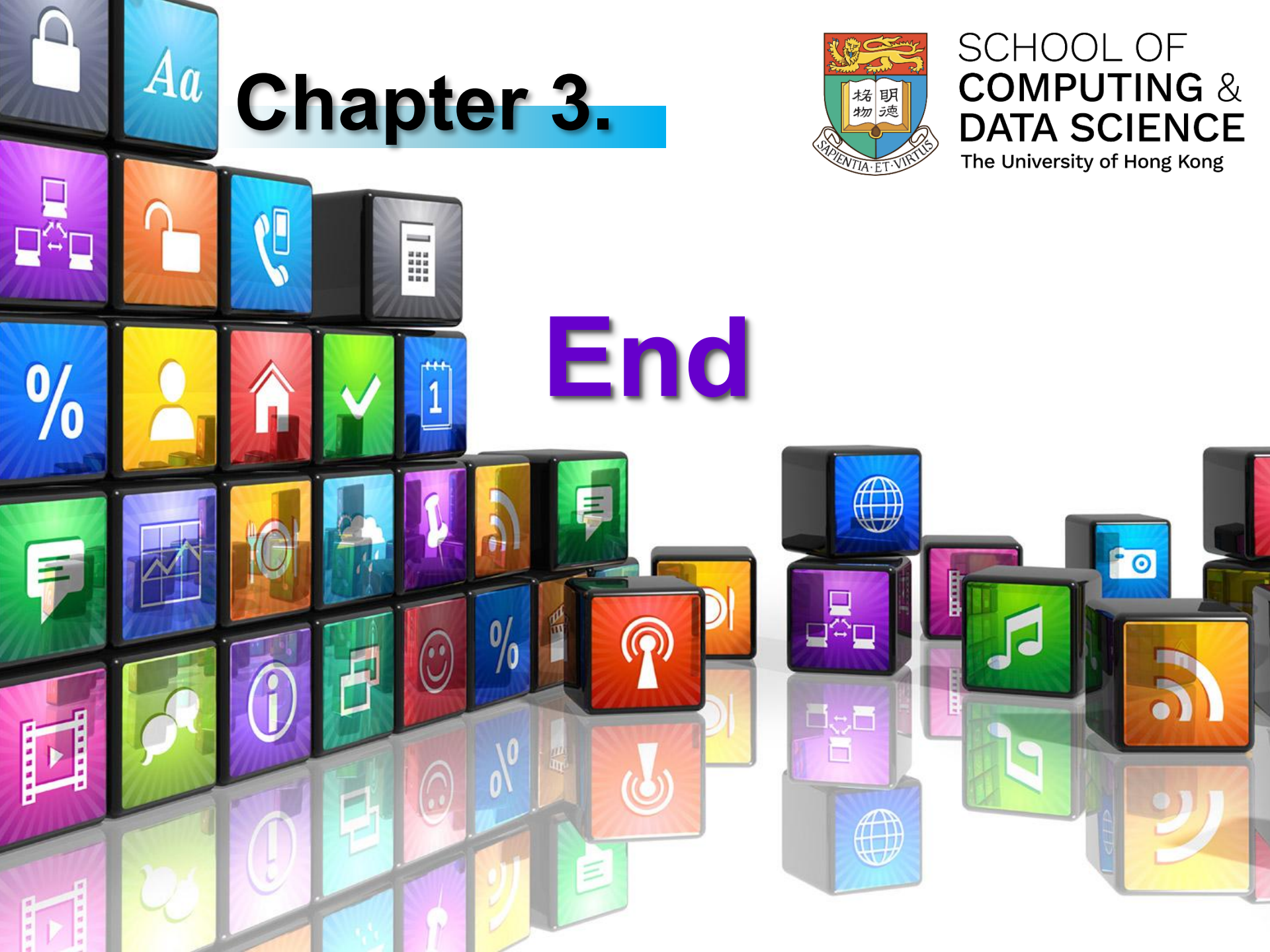
- Mini Program can be regarded as a hybrid APP. Let's take WeChat Mini Program as an example.
  - Bottom layer: a container provided by Native
  - Upper layer: WXML (WeChat Markup Language) & WXS (WeChat Script) & JS provided by WeChat for business development (WXML and WXS are developed from HTML and CSS).
  - Very similar to a browser and server-side architecture

Mini Program

- Development tools are available for the development. An example is WeChat DevTools (https://developers.weixin.qq.com/miniprogram/en/dev/devtools/devtools.html).



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